DMITRII GUSEV

dmitrii@gusev.tech | +1 604 440 1250 | linkedin.com/in/dmitriigusev | github.com/amidg

TECHNICAL SKILLS

Programming & Scripting: C++, C, Python, Bash, MATLAB & Simulink, TwinCAT, TwinSAFE

Build & CI/CD: Make, CMake, GCC, GDB, Clang, Colcon, Git, Google Test, GitLab & GitHub CI/CD

Development: Object-Oriented Programming, Agile, Test-Driven Development, Jira, JAMA, Claude Code

Frameworks: FastDDS & ConnextRTI DDS, ROS2, ros2_control, MoveIt, Gazebo, OpenCV, TensorRT, Ollama

Platforms: Linux (Ubuntu & RHEL), Docker & Podman, Kubernetes, QEMU & KVM (+VFIO), FreeRTOS

Hardware: NVIDIA Jetson, ESP32, STM32, EtherCAT, CAN, Modbus, RS232, RS485

Electrical & Lab Equipment: Altium, LTSpice, Soldering, Multimeter, Oscilloscope, Logic Analyzer

WORK EXPERIENCE

Sanctuary AI — ***Senior Robotics Software Engineer*** — ***Aug. 2024 – Present***

Vancouver, BC

- Leading low-level C++ software development for the next generation of an autonomous humanoid robot platform: middleware, hardware interfaces design (EtherCAT, CAN, RS485, GMSL2) and system integration with machine learning and perception stack
- Re-architected entire company's compute platform achieving 30% reduction in real-time CPU scheduling latency and 40% improvement in AI inference throughput while reducing power consumption by up to 90%
- Saved $2M/yr in licensing costs by initiating and coordinating transition to FastDDS middleware across entire stack
- Developed an internal FastDDS-based C++ library with a focus on shared memory communication and thread-safe operation within Real-Time Operating Systems environment
- Implemented high-performance motor controls in C++ using Acontis EtherCAT driver and FastDDS middleware on a PREEMPT_RT-enabled Real-Time Linux PC focusing on achieving deterministic operation and functional safety at 1KHz control loop frequency
- Architected company-wide monorepo consolidating controls, real-time and teleoperation software stack for all robot platforms reducing physical AI embodiment bring-up time from 3 months to 60 hours
- Actively doing external vendor management across North America and China

Industrial Next (YC W22) — ***Founding Software Engineer*** — ***Sept. 2022 – June 2024***

San Francisco, CA

- Designed multiple robotics and computer vision projects across multiple production lines using ROS2 and MoveIt
- Achieved sub-mm precision of a robotics task by introducing a load-cell controlled over CAN by an ESP32 running FreeRTOS
- Architected and designed IEC61131 compliant digital IO module PCB using ESP32 and multiple MAX14906 via SPI daisy-chain
- Supported bring and debugging of a custom Jetson carrier PCBA: power sequencing and Image Signal Processor
- Upgraded company's C++ camera software stack with the dual imager GStreamer pipeline support
- Developed motion planning for the time-sensitive robot arm applications using MoveIt2 and C++; Achieved the desired cycle time of a motion planning task using trajectory pre-caching and optimization via Curvature Corrected Moving Average
- Provided production critical bugfixes for ROS2 Foxy in C++ and ROS2 Python state-machine library
- Implemented driver written in Python for the Advantech industrial IO and PWM modules over Modbus-TCP
- Led the implementation of the robot digital twin and coordinated development with machine learning team
- Contributed to the company's core software IP implementing multithreaded code for data processing and error handling
- Executed multimillion-dollar deployment of the CV/ML application at Dell laptop assembly line using Kubernetes
- Collaborated with stakeholders and clients across Asia and the US responding to changing business and technical conditions

Resume showcases only relevant work experience. Full portfolio and references are available upon request.

DMITRII GUSEV

dmitrii@gusev.tech | +1 604 440 1250 | linkedin.com/in/dmitriigusev | github.com/amidg

Tesla — ***Manufacturing Controls Software Engineer*** — ***Sept. 2021 – Sept. 2022***
Toronto, ON / Austin, TX

- Designed TwinCAT controls programs for the 4680-battery terminal assembly machines at Giga Texas and Berlin
- Participated in design reviews and applied Agile development methodologies to controls development workflows
- Led the development of the laser marking systems for the 4680-battery manufacturing lines achieving 300 parts/min
- Innovated internal tools in Python automating PLC codebase maintenance saving 20+ hrs./engineer/month
- Introduced MATLAB-based tool for one-click torque calibration of the Beckhoff VFD drives installed on the factory line
- Commissioned equipment for the 4680-battery lines achieving desired performance within strict deadlines
- Recruited and mentored software engineering interns fostering their professional growth

Tesla — ***Engineer Intern*** — ***Jan. 2019 – Dec. 2020***
Palo Alto, CA

- Designed a PCB with ESP32 and CAN (MCP2515) for the addressable safety light control using FreeRTOS
- Developed C++ ROS nodes for the Tesla AGV platform: line sensing, safety lights and fleet management integration
- Introduced a bugfix for the C++ firmware of the Model 3 FPD Link III to HDMI board touch screen behavior
- Programmed firmware of a power distribution board used on the Tesla AGV platform
- Implemented QR code detection using OpenCV at Fremont Factory and reduced storage conveyor downtime by 70%
- Established a new computer vision process at the factory allowing to save up to $5000 per incoming delivery truck
- Promptly prototyped and deployed an autonomous traffic light system for the human-to-AGV interaction at Giga Nevada using Python and Raspberry Pi
- Coordinated third-party vendors on the implementation of the truck yard tracking using Zigbee-based hardware
- Contributed to the internal Beckhoff PLC programming methodology and programmed a battery sorting machine
- Engineered a PCBA assembly for the motor controller cooling for the first generation Tesla Optimus robot

Electra Meccanica — ***Vehicle Engineer Intern*** — ***Apr. 2018 – Sept. 2018***
Vancouver, BC

- Designed FMVSS108 certification test jig using Arduino Mega2560 implementing relay control and SD card logging
- Engineered a PCBA assembly for “Solo” vehicle’s windshield controls passing FMVSS103 and FMVSS104 certifications
- Developed a vehicle’s harness test jig using Arduino Mega2560 increasing OEE of the vehicle manufacturing line
- Performed >30 hours of dyno testing and optimized the motor controller and BMS firmware behavior resulting in up to 10% NEDC range increase
- Created a PCB to display battery State of Charge read over CAN on an analog fuel gauge using I2C potentiometer
- Integrated SAI J1772-compliant charging into the Intermeccanica eSpeedster prototype enabling charging capabilities

PERSONAL ENGINEERING PROJECTS

Autonomous Mobile Robot — ***Embedded Systems Engineer*** — ***Aug. 2019 – Aug. 2021***
Vancouver, BC

- Architected entire electrical system for the Capstone Autonomous Mobile Robot project targeting modularity and scalability within automotive manufacturing environment
- Designed PCBs in Altium: BLDC Motor Driver, Safety Controller, Battery Pack BMS, and Power Delivery Controller
- Engineered circuits for Ethernet PHYs, USB-OTG, current and voltage monitoring via ADC and power rail sequencing
- Developed FreeRTOS-based firmware for ESP32-based PCBs and implemented ROS Serial communication via UART

EDUCATION

Simon Fraser University — ***2021***
Vancouver, BC

Bachelor of Applied Science. Mechatronic Systems Engineering. Dean’s Honour Roll (2021)
Team “Phantom”: SFU Formula SAE Electric – Lead Battery Pack Engineer (2016 - 2019)

Resume showcases only relevant work experience. Full portfolio and references are available upon request.